CrowdScope: A Modular Framework for Crowd-State Analysis and Uncertainty-Aware Resource Planning

Surya Maheswaran
Btech. (Hons), SoCSE
RV University
Bengaluru, India
suryam.btech23@rvu.edu.in

Yashaswini Pawar
Btech. (Hons), SoCSE
RV University
Bengaluru, India
yashaswinip.btech23@rvu.edu.in

Sohan Kori
Btech. (Hons), SoCSE
RV University
Bengaluru, India
sohank.btech23@rvu.edu.in

Dr. Seema B Hegde
SoCSE
RV University
Bengaluru, India
seemabh@rvu.edu.in

***Abstract*—Crowd management requires spatial measurements, an understanding of population heterogeneity, and an explicit connection between uncertain observations and decisions. This paper presents CrowdScope, a modular trajectory-to-decision framework integrating geometry-aware crowd measurement, movement-based wheelchair-associated classification, probability calibration, zone composition estimation, and capacity-constrained resource planning. The detector uses four seconds of recorded planar movement without images, height, or participant identifiers. An exploratory comparison of fifteen models and controls on eight SiME trials identifies full-feature logistic regression as the leading ranking model, with average precision 0.907. Nested calibration excluding the target wheelchair participant and recording trial reduces Brier loss from 0.0362 to 0.0210; calibrated precision and recall are 84.3% and 83.6%. A separate coherent whole-trial replay connects the modules at 48 checkpoints, reducing wheelchair-count mean absolute error in nonempty zones from 0.602 with a population prior to 0.452 people per zone. This replay includes recurring participants and is not unseen-person validation. The default hypothetical resource-allocation outcome does not improve, demonstrating that better crowd estimates do not necessarily produce better decisions. We describe how managers can define observation zones, inspect density and composition uncertainty, and compare resource assumptions. Limited participant diversity, tracking effects, detection coverage, and unobserved service demand constrain deployment. The contribution is an auditable integrated framework with bounded empirical evidence, rather than a universal demographic recogniser or evacuation predictor.**

***Index Terms*—Conditional probability, crowd analysis, decision support, mobility recognition, resource allocation, uncertainty quantification.**

I. Introduction

Crowd managers need to know where people accumulate, how movement changes, and which locations may require attention. Occupancy alone is insufficient: the same number of people can move freely across an open area or form a slow queue at a narrow passage. Experiments involving people with different mobility characteristics demonstrate changes in spacing, speed adaptation, and interaction [1], [2]. Accessibility-oriented planning therefore benefits from both spatial crowd state and information about population composition.

However, observing slow movement does not establish wheelchair use, and wheelchair use does not establish a request for assistance. Congestion, waiting, route geometry, and tracking noise can produce similar movement summaries. Even accurate probabilities may leave the preferred resource allocation unchanged. A useful crowd-analysis system should preserve these distinctions instead of translating a hard demographic label directly into a staffing recommendation.

We present CrowdScope, a modular framework connecting time-stamped trajectories and walkable geometry to crowd measurements, calibrated wheelchair-associated probabilities, uncertain zone composition, and a configurable resource decision. Its implemented modules support recorded-data analysis and streaming ingestion. A local demonstration replays actual SiME movement; a separate synthetic application supports controlled what-if analysis. Neither interface requires CCTV processing.

The work makes three contributions. First, it implements a causal trajectory-to-decision interface that distinguishes measured occupancy, inferred membership, missing evidence, and assumed service demand. Second, it evaluates a position-only recognition component using person/trial exclusions and nested calibration, alongside simpler, ensemble, and neural alternatives. Third, it connects the calibrated estimates to an auditable whole-crowd replay and measures both composition error and decision outcomes. The mathematical methods are established; the contribution lies in their integration and the explicit assessment of where improved information does and does not help.

The architecture can accommodate different crowd-management applications through declared observation regions and decision modules. The current evidence is narrower: controlled bottleneck recordings, seven documented wheelchair participants, and a hypothetical static allocation task. We do not claim validated recognition of children or older adults, general transfer across tracking systems, or operational evacuation-time prediction.

II. Literature Review and Related Work

A. Search strategy and selection

The focused review combines the existing evidence register with targeted searches of primary publisher, institu-

tional, software-documentation, and author-repository sources. Recorded query families included *crowd trajectory disability wheelchair classification probabilistic resource allocation evacuation mobility uncertainty*, *Bayesian crowd evacuation resource allocation uncertainty mobility impairment*, and combinations of *crowd*, *composition*, *uncertainty*, and *planning*. Follow-up searches examined SiME trajectory extraction and probability calibration. No publication-year restriction was imposed; the bibliography includes foundational work through 2026.

We retained work on heterogeneous movement, trajectory measurement, relevant recognition modalities, calibration, and resource decisions. DOI/repository identifiers were used to compare records; unresolved access limits remain in the register. No authenticated Scopus or Web of Science export was obtained. The eighteen references constitute a focused bibliography, not a claim of eighteen systematically screened PRISMA studies.

B. Measurement, recognition, and planning

SiME research establishes heterogeneous movement effects [1], [2]; the underlying tracked coordinates represent head positions projected into the ground plane [3]. Density depends on measurement support, with alternatives such as Voronoi estimation offering different scatter properties [4]. Schumann *et al.* describe hybrid tracking for wheelchair users and explicitly discuss fluctuations associated with walking head sway [5]. Consequently, a lateral-motion feature is plausible but cannot be claimed as a newly discovered physical mechanism, and recording effects require attention.

Lin *et al.* integrate LiDAR-based subclassification and arrival prediction [6]. Vásquez *et al.* recognise mobility aids using depth imagery and probabilistic tracking [7]; Woo *et al.* provide smartphone/watch activity data [8]. These modalities differ from position-only histories. We use independently documented wheelchair associations rather than assigning labels from the same speed rule the model is asked to learn.

Probability calibration is an established mapping from classifier scores to empirical event frequencies [9]. Parameter uncertainty in pedestrian models has also been studied through Bayesian calibration [10]. Rescue-guide optimisation [11] and stochastic evacuation recruitment/routing [12] already connect uncertainty with decisions. Table I positions our scope relative to these predecessors. Neither Bayesian inference nor uncertainty-aware allocation is introduced here as a new algorithm.

The research opportunity is a transparent end-to-end comparison separating recognition, probability quality, composition estimation, and decision value. The review does not establish a verified first framework or exhaustive absence of related integrations.

III. Research Problem, Objectives and Hypotheses

Given trajectories $\mathcal{X}_{\leq t}$ and walkable geometry G, the problem is to produce a crowd state S_t and an application-specific decision a_t without using future observations or treating uncertain profile estimates as known needs. The objectives are to measure local crowd conditions, recognise a supported mobility-associated class, calibrate its scores, estimate zone composition, and evaluate downstream allocation.

RQ1 asks whether movement features distinguish documented wheelchair-associated tracks better than speed-only and population baselines. RQ2 asks whether calibration improves probability quality and zone counts. RQ3 asks whether the resulting estimates improve a fixed resource decision and how managers can interpret their limits.

The recognition and calibration extensions are exploratory: data inspection preceded their protocols, and the model-family comparison informed selection. Their directional propositions are that full movement features outperform speed-only ranking and that calibration reduces held-out Brier loss. We report their observed direction without treating repeated tracks as independent people or asserting confirmatory hypothesis rejection.

The retained controlled planning experiment has a separate frozen hypothesis, $H_1 : \mathbb{E}[\Delta] < 0$ versus $H_0 : \mathbb{E}[\Delta] \geq 0$, where Δ is probabilistic minus hard-posterior unmet requests under its base synthetic mechanism. Its original comparator, seed blocks, and analysis are unchanged. This historical hypothesis does not become evidence for a new recogniser merely because that recogniser is attached to the framework.

Independent variables include feature/information representation and calibration; dependent variables include average precision (AP), probability losses, count error, and conditional service shortfall. Geometry, observation horizon, capacity, and model settings are controlled. Congestion, assistance, repeated identity, tracking quality, and population shift are potential confounders.

IV. Research Methodology

A. Architecture and causal crowd measurement

We use a quantitative design-science approach: implement the framework, verify its interfaces, and evaluate distinct evidence streams. Fig. 1 shows the workflow. Input records contain a run-local ID, timestamp in seconds, and planar position in metres. Walkable polygons and analysis zones define spatial support. The state interface exposes occupancy, usable area, density, past motion, observation coverage, membership probabilities, their provenance, and the assumptions needed by a decision module.

For zone z of usable area A_z, occupancy and density are

$$N_z(t) = \sum_i \mathbf{1}\{x_i(t) \in z\}, \qquad \rho_z(t) = N_z(t)/A_z. \quad (1)$$

Polygon areas exclude obstacles. Zones have disjoint interiors; shared boundaries belong to the lowest matching zone ID. Exact observed timestamps determine membership, absent tracks are excluded, and geometry conflicts are reported rather than clipped away. Missing observations are not automatically classified as exits.

Movement summaries use only the preceding two seconds. Mean speed is path length divided by valid contiguous duration; gaps break continuity and a lone observation has

TABLE I
SELECTED PREDECESSORS AND THE PRESENT EVALUATION SCOPE

Study	Data and method	Relevant contribution	Distinction from CrowdScope
Geoerg et al. [1], [2]	Controlled heterogeneous passage experiments	Empirical changes in movement and interaction	Our detector and resource outcomes require separate evaluation
Schumann et al. [5]	Camera/IMU wheelchair tracking	Occlusion handling and recorded movement fluctuations	Tracking differences may confound our position-only classification
Lin et al. [6]	LiDAR subclassification and arrival models	Connects recognition and time prediction	We study calibrated composition and conditional zone decisions
von Schantz / Moug et al. [11], [12]	Scenario-based guide or transport optimisation	Uncertainty-aware resource decisions	Their dynamic evacuation tasks differ from our static service abstraction
Present study	Real trajectories, detector comparison, nested calibration and replay	Explicit observation–inference–decision interface	Small repeated-participant cohort; real service demand unobserved

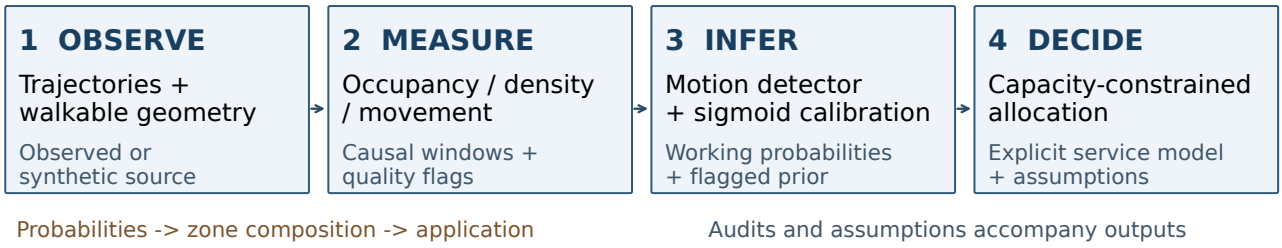


Fig. 1. Implemented crowd-management pipeline. Recorded positions support measurements and a calibrated movement detector. Missing detector evidence retains a flagged prior. Service requests remain an explicitly configured model; optional simulation is a separate diagnostic branch.

unknown speed. Stopping uses 0.1 m/s, with explicit stopped-time and low-mean-speed summaries. The framework retains coverage and missing-motion flags. Invalid units, duplicate observations, overlapping zones, and nonfinite values produce explicit errors.

B. Audited labels and fixed observation windows

The broader archived SiME collection contains fourteen trials and 1,151 person-trial tracks [13]. Recognition is restricted to trials 501–508. Each run's metadata declares seven participants with disabilities, and its trajectory header maps all seven documented wheelchair IDs 34–40 to present tracks. Observed total counts match metadata in these eight trials. Under the explicit completeness assumption, the remaining tracks form a non-wheelchair roster complement; this rule is not transferred indiscriminately to other SiME groups.

We take one four-second window ending at the first observed forward crossing of $x = -1$ m. This event can trigger online inference without future frames. A valid window has 101 consecutive samples at 25 Hz. One wheelchair-associated track lacks that crossing, leaving 711 eligible windows: 55 positive person-trial observations of seven people and 656 controls. Most controls lack verified persistent identities.

The 54 tabular features describe speed and absolute lateral velocity at 0.04, 0.2, 0.4, and 0.8 s displacement scales; stopping, forward motion, variability, straightness, lateral reversals/acceleration; and detrended lateral spectral power in 0.25–1, 1–3, and 3–8 Hz bands. Neural sequence models receive the 100 planar velocity vectors. Height, absolute position, frame/trial number, identity, and future passage duration are excluded as predictors. The input remains passage-aligned; new orientations require an adapter and validation.

C. Model comparison and nested calibration

Fifteen fixed models/controls are compared: a population prior; speed-only, lateral-only, coarse-scale, and full logistic models; Gaussian naive Bayes; kNN; RBF SVM; random forest; Extra Trees; histogram gradient boosting; feature MLP; temporal CNN; GRU; and shuffled-training-label logistic control. Standardisation is fitted within training only. Settings are modest and fixed: full logistic $C = 0.2$ with balanced classes; forests use 120 trees and minimum leaf size three; boosting uses 100 iterations and seven leaves. Neural models use 60 epochs, one deterministic seed per fold, and small fixed architectures. The repository records all settings; no exhaustive tuning or universal architecture ranking is claimed.

Each crossed outer fold excludes one trial and one target wheelchair participant from every training record. Test data contain that participant in the held trial plus its controls; the other six wheelchair people in that trial are not tested in that fold. Known mapped test-control identities are also excluded from training. There are 55 eligible folds. Each

positive is tested once; controls receive six or seven out-of-trial predictions. Inverse-count control weights preserve one effective observation per control. Unknown repeat controls cannot be excluded, so only the target wheelchair-person exclusion is guaranteed. This structure follows the need to respect grouped dependence [14].

AP is the descriptive family-selection criterion, not classification accuracy. The leading full logistic model produces score s_i. A regularised sigmoid calibrator [9] estimates

$$p_i = \mathbb{P}(W_i = 1 \mid s_i) \approx \sigma(\alpha s_i + \beta), \quad \sigma(u) = (1 + e^{-u})^{-1}. \tag{2}$$

Within each outer training set, inner crossed person/trial predictions fit this calibrator at natural prevalence, with $C = 1$. Across outer folds, 2,268 inner fits exclude all outer-test observations from both fitting stages. The base model is then fitted to outer training and scored on outer test. The outer comparison remains exploratory because prior data inspection and family selection used the same study material.

We report weighted Brier loss [15], log loss, AP, ROC-AUC, preset-threshold precision/recall, and fixed-bin reliability. Bins have edges 0, 0.1, 0.2, 0.4, 0.6, 0.8, 1. Brier loss reflects discrimination as well as calibration, so its reduction alone is insufficient. The final prototype refits the base on all development data and calibrates using original crossed out-of-fold scores; validation results are never computed from that all-data refit.

D. Composition, missing evidence, and allocation

The streaming detector is invoked only at the validated crossing with an uninterrupted four-second history. Its probability is cached as uncertain stable membership information. Before eligibility or with inadequate history, a flagged training prior is retained. The same movement evidence is not subjected to a second Bayesian update. Expected wheelchair-associated count is

$$\mathbb{E}[N_{z,W} \mid \mathcal{X}_{\le t}] = \sum_{i \in z} p_i. \tag{3}$$

A working Poisson-binomial distribution provides variance $\sum_i p_i(1 - p_i)$ and 90% count intervals. Shared model errors, crowd selection, and correlated memberships can invalidate independence. A fractional expectation is not a hard count or medical diagnosis.

Membership and service demand are separate variables. For request indicator R_i and configured rates r_0, r_1,

$$q_i = \mathbb{P}(R_i = 1 \mid \mathcal{X}_{\le t}) = r_0 + (r_1 - r_0)p_i. \tag{4}$$

The rates are assumed, not estimated from actual requests. With zone demand $D_z = \sum_{i \in z} R_i$, capacity k, and budget B, the application solves

$$\min_{a_z \in \mathbb{Z}_{\ge 0},\, \sum_z a_z = B} \mathbb{E}\left[\sum_z (D_z - k a_z)_+\right]. \tag{5}$$

Independent requests yield Poisson-binomial demand distributions. Four zones and six units permit exhaustive evaluation of 84 allocations; ties within 10^{-12} use lexicographic order. Resources are interchangeable and prepositionable. No travel time, queue dynamics, clinical staffing standard, or actual assistance graph is represented.

E. Whole-crowd replay and controlled planning evidence

A coherent full-roster replay needs one model for the entire held trial. Excluding all seven wheelchair identities simultaneously would leave no wheelchair training examples. We therefore use a *separate* leave-one-trial-out evaluation, calibrated by seven inner trial folds per outer trial (56 additional fits). People may recur between training and test. These results must not be presented as unseen-person whole-crowd validation.

Each trial's walkable polygon is intersected with $[-6, 2.4] \times [-1.65, 1.65]$ m and split at $x = -6, -3, -1, 0, 2.4$. The 25 Hz stream emits one-second replay states; fixed checkpoints are 10, 20, 30, 40, 50, and 60 s. We compare prior-only, raw, calibrated, and thresholded composition estimates. Labels enter only the evaluation of counts and a known-class resource reference. Planning uses $B = 6$, $k = 2$, and assumed $r_0 = 0.1, r_1 = 0.4$; expected loss is evaluated under those label-conditioned hypothetical request rates, not observed human requests.

The earlier integration study of fourteen trials and 42 checkpoints is retained as measurement evidence. A separate frozen synthetic planning experiment retains eighty occupants, fourteen conditions, sixty seed blocks, ten replicas per condition, and seven policies (58,800 records). Its supplied binary observations follow a declared symmetric-error channel, with Bayesian posteriors rather than a trained detector. Replicas are averaged within seed; the original paired percentile bootstrap uses 10,000 resamples [16]. It is not rerun or relabelled as real-detector validation.

F. Verification and ethics

Forty-six automated tests cover units, gaps, future invariance, polygon areas, probability recursion, exact allocation, prior fallback, cached estimates, and absence handling. Independent audits check hashes, nested exclusions, probability metrics, temporal provenance, and conservation. One hundred recorded-state fixtures compare Python and browser allocation results.

JuPedSim 1.4.2 [17] with collision-free speed dynamics [18] remains an optional diagnostic backend. Small movement fixtures pass, while larger cases retain jams, wall failures, and timestep sensitivity. Simulation neither supplies recognition labels nor replaces independent wheelchair participants. No new participants were recruited; human trajectories, per-person scores, local replay files and fitted recognition weights remain excluded from the GitHub update pending redistribution confirmation.

V. Results

A. Detection and probability calibration

Fig. 2 compares all models/controls and reliability. Full logistic leads on crossed-holdout AP (0.9074), followed by

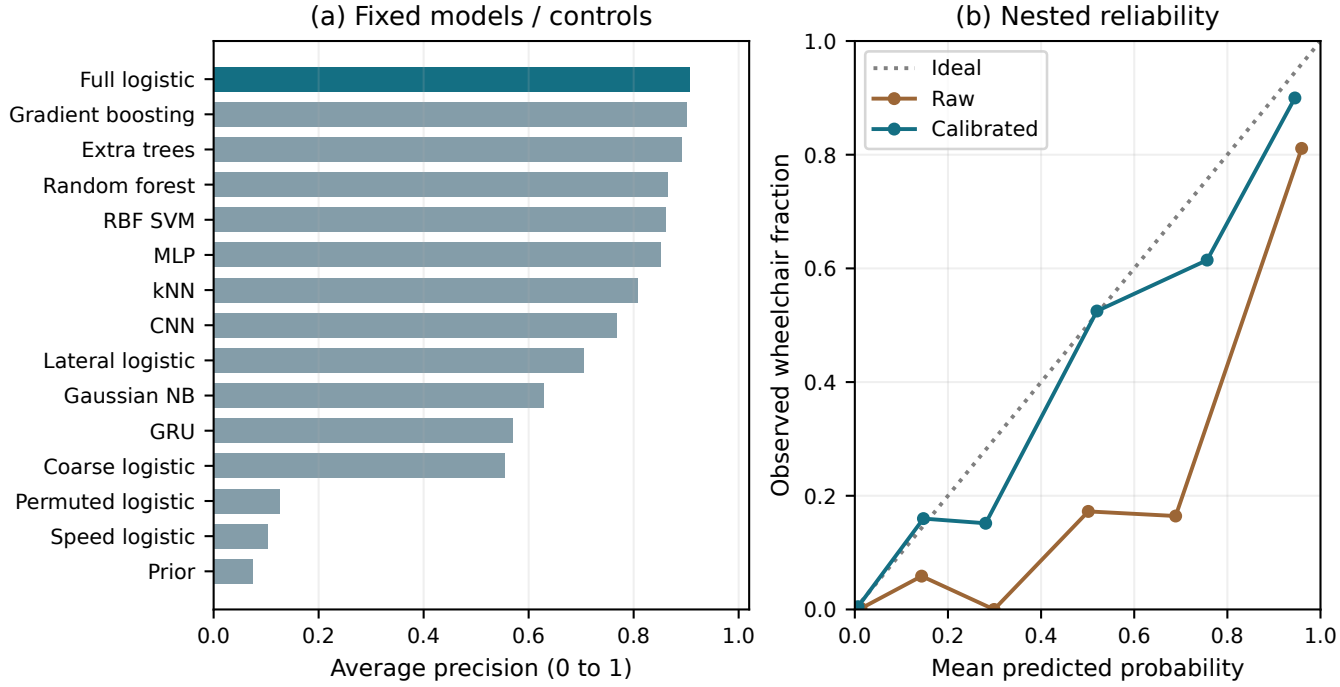


Fig. 2. Exploratory person/trial holdouts. Left: AP for the fifteen fixed raw models/controls; AP is not accuracy. Right: nested sigmoid calibration improves fixed-bin reliability relative to raw full-logistic scores. Repeated control predictions are inverse-count weighted; the cohort contains only seven wheelchair people.

TABLE II
PERSON/TRIAL-HELD PROBABILITY EVALUATION (WEIGHTED)

Input	Brier ↓	Log loss ↓	AP ↑
Training prior	0.07138	0.27232	0.07446
Raw full logistic	0.03623	0.12322	0.90745
Calibrated logistic	0.02100	0.07312	0.90836

TABLE III
WHOLE-TRIAL REPLAY: COMPOSITION AND CONDITIONAL PLANNING

Input	Count MAE[a]	Count bias[a]	Mean loss[b]
Population prior	0.6019	−0.0299	0.085408
Raw detector	0.5156	0.0544	0.085494
Calibrated detector	0.4520	−0.0408	0.085494
Hard calibrated labels	0.5722	−0.5508	0.084004
Known-class reference	—	—	0.074723

[a]People per nonempty zone. [b]Expected unmet requests under assumed independent rates 0.1/0.4, averaged over 48 checkpoints; no real service outcomes.

boosting (0.9014) and Extra Trees (0.8917). MLP, CNN, and GRU obtain 0.8517, 0.7687, and 0.5693. Speed-only AP is 0.1024; lateral-only AP is 0.7047. Temporal and lateral information therefore offer a useful candidate signal in these recordings, while speed alone is weak. At preset thresholds, random forest has the highest observed F1 (0.8270); "best" depends on the selected criterion.

Table II reports probability scores. Calibration reduces Brier loss by 42.0% and log loss from 0.1232 to 0.0731. Fixed-bin expected calibration error decreases from 0.0572 to 0.0082. Calibrated precision and recall at 0.5 are 84.3% and 83.6%; raw values are 64.2% and 94.5%. Mean calibrated probability is 0.0845 versus observed prevalence 0.0774. Reliability improves without becoming exact.

The shuffled-label sanity control has AP 0.1271, substantially below the leading model but not exactly the prevalence baseline. A single permutation is not a formal permutation test. Coarse-time features alone obtain AP 0.5556; because this ablation also changes feature content, it cannot isolate head sway as the sole cause. Neural training scores often exceed holdout scores, and the fixed neural configurations do not dominate simpler models. No further family tuning follows these outcomes.

B. *Connected crowd-state and planning outcomes*

The coherent held-trial stream produces 759 replay states. All 711 eligible online detector outputs agree with corresponding offline predictions. All 48 fixed checkpoints integrate without geometry mismatch; counts are conserved across 192 zone records, of which 187 are nonempty. Table III shows a roughly 25% reduction in nonempty-zone count MAE relative to the population prior. These are trial-held, recurring-participant results, distinct from the preceding person/trial-held probability evaluation.

Mean detector coverage is only 21.5% across nonempty checkpoints: most approaching occupants have not reached the validated trigger. Their probabilities remain flagged priors. Working 90% intervals cover the recorded count in 97.9% of

nonempty zones, suggesting conservative intervals here rather than established nominal coverage elsewhere.

The calibrated resource policy does not outperform prior-only or hard-label allocation under the default request model. Raw and calibrated policies have identical mean loss despite different composition estimates. Thus the positive recognition/count results do not establish a service benefit. The known-class reference also retains unmet demand because class knowledge is not perfect demand knowledge.

C. Retained controlled and simulation findings

In the original synthetic base condition, probabilistic planning reduces mean unmet requests by 0.255 versus the frozen weak hard-posterior comparator (paired 95% interval $[-0.390, -0.127]$). However, it matches hard-observed allocation in 599/600 cases and has no base advantage over that stronger control. Small secondary gains depend on zone imbalance and demand contrast. All conditions remain available in the repository.

The earlier 42-checkpoint measurement integration remains successful, with its metadata discrepancy retained. Diagnostic simulations still fail to establish general timestep-stable clearance. Conditional matched replays improve on free flight but use observed inlet events and cannot be described as evacuation forecasts. These separate findings prevent the new detector from being used to retroactively strengthen unsupported simulation claims.

VI. Discussion

A. How managers can use the framework

A user first defines the observable walkable region and management zones, then supplies timestamped trajectories through the CSV adapter or streaming API. Units, sampling, passage alignment, and geometry must be checked. Density and past speed identify concentrations and slow-moving regions; coverage flags distinguish reliable measurements from missing history. Users can compare snapshots over time without interpreting a disappeared track as a completed evacuation.

The composition layer adds expected wheelchair-associated counts and uncertainty, not definitive identities. At 40 s in trial 501, for example, 28 occupants produce a peak zone density of 1.62 people/m^2 and an expected wheelchair-associated count of 3.43, with a working interval of 1–6. Only five occupants have detector evidence at that moment; the remaining estimates use priors. A manager should inspect that coverage before interpreting the total as a detailed accessibility assessment.

For resource exploration, users configure available units, per-unit capacity, and service-specific request assumptions. The planner compares where capacity could be positioned, while the interface exposes each zone's density, expected composition, evidence source, and proposed allocation. This can support discussion of staff observation points or accessibility-support readiness, followed by human verification of actual needs. It does not justify treating every predicted wheelchair user as needing assistance.

Other management tasks can consume the same state through different decision modules. Occupancy monitoring needs appropriate local thresholds; route guidance additionally needs travel/queue models and validated routing policies; dynamic dispatch needs service times, travel and reassignment costs. The current framework provides inputs and interfaces for those extensions, not validated recommendations for every use case. The local replay illustrates the workflow, while the shareable synthetic demo explores assumptions separately.

B. What the evidence does and does not establish

The detector closes an earlier gap between movement observations and composition estimates. Its stronger results relative to speed-only controls support further study, and nested calibration reduces the overstatement produced by balanced classifier scores. However, the calibrated posterior should be passed directly to composition estimation; reapplying Bayes to the same movement evidence would count it twice.

More accurate state estimates do not guarantee better decisions. Resource allocation is discrete, demand remains assumed, and multiple probability representations may select the same action. CrowdScope therefore reports recognition scores, probability quality, count errors, decision changes, and losses separately. The negative default planning result is part of the contribution rather than a reason to select favourable service rates after inspecting outcomes.

C. Threats to validity

Seven wheelchair participants provide limited diversity, and many control identities are unknown. The recognition cohort depends on complete documented rosters. Data exploration and family selection precede the nested calibration comparison; further neural tuning on these same observations would increase selection optimism. Fixed epochs and one seed per fold do not establish an optimal architecture. No untouched external dataset or independent-participant confidence interval is available.

Tracked head motion can reflect sway, assistance, occlusion, smoothing, and camera processing [5]; identical processing across classes is not verified. The crossing-aligned detector is not validated for arbitrary histories or scenes. Population/recording shifts can break calibration, and the low pre-trigger coverage limits early composition analysis. Working independent-count intervals omit shared model error.

The full-roster evaluation intentionally permits recurring participants and cannot be conflated with individual person/trial exclusion. Its zone checkpoints are dependent. Demand is not empirically annotated; the allocator omits time and movement costs. The focused literature review cannot verify first-of-kind novelty, and simulation convergence failures prohibit operational clearance claims.

VII. Conclusion and Future Work

CrowdScope now connects measured trajectories to calibrated wheelchair-associated probabilities, local crowd composition, and an explicit planning module. Exploratory person/trial holdouts favour full logistic movement features over

speed-only recognition; nested calibration improves probability quality. Separate whole-trial replay improves composition-count estimates, but the default hypothetical allocation outcome does not improve. The framework makes this distinction visible to users rather than equating detection accuracy with management effectiveness.

The next priorities are independent wheelchair participants and tracking systems, earlier observation windows, abstention, population-shift testing, and correlated uncertainty. Service-demand annotations and travel/service-time models are needed before operational resource evaluation. Child/age recognition requires separate labels and validation. Any synthetic augmentation must improve unchanged real holdouts rather than encode an invented class signature. Simulation requires geometry/contact and timestep convergence before supporting passage forecasts. These extensions can broaden the modular framework through additional evidence.

Declarations and availability. No external funding was received and no new participants were recruited. Code, the paper, aggregate evaluation results, model documentation and demo interfaces are provided at https://github.com/surya070/crowdscope. The repository is private and requires access permission. Raw/derived participant trajectories, individual predictions, fitted recognition weights and the recorded replay package are excluded pending redistribution confirmation; the synthetic demo remains self-contained. The local recorded replay uses held-trial models, while the all-development refit is identified separately. Author declarations and venue-specific requirements should be confirmed before submission.

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